higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING FABRICATION: SHEET METAL WORK
NQF LEVEL 3

(6031023)

21 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 8 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use only BLUE or BLACK ink.
5. Write neatly and legibly.

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QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 … is an alloy steel.

A Cast iron
B Tool steel
C Stainless steel
D Aluminum

1.2 The … chisel is used to cut oil grooves in a solid bearing.

A flat
B cross-cut
C diamond point
D round-nose

1.3 The … is an indirect measuring instrument that can be used to measure the inside diameter of a round object.

A divider
B inside caliper
C jenny caliper
D outside caliper

1.4 The volume of a round tank can be calculated by the following formula:

A $V = \pi r^2$
B $V = \pi D^2$
C $V = \frac{\pi D^2}{4}$
D $V = \pi r^2 \times h$

1.5 … is a technique used to form various sheet metal components.

A Welding
B Bolting
C Laser cutting
D Brazing

1.6 … is a temporary method of assembling.

A Bolting
B Arc welding
C GMAW welding
D TIG welding

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1.7 The size of a gas welding torch nozzle is determined by …

A the type of material.
B the hardness of the material.
C the colour of the material.
D the thickness of the material.

1.8 To crack the cylinder valve means to …

A open the valve.
B close the valve.
C break the valve seal.
D rapidly open and close the valve.

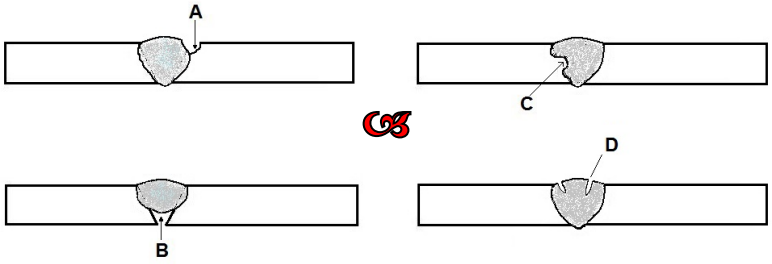
1.9 Which ONE of the following statements is *incorrect* with regard to welding safety?

A Compressed air can replace the oxygen supply in a welding process.
B Never use compressed air to clean clothing.
C Store and use acetylene cylinders with the valve end up.
D Use valve protection caps on cylinders when moving them.

1.10 ... is classified as resistance welding.

A Swaging
B Corner weld
C Spot welding
D Butt welding

1.11 Which drawing of the following indicates insufficient fusion?



1.12 Grease and oil can easily be removed from the welding area by using …

A a steel brush.
B an angle grinder.
C an oxy-acetylene flame.
D degreasing fluid.

1.13 A scale of 1:100 means …

A 1 Measuring unit on the product represents 100 measuring units on the drawing.
B 1 Measuring unit on the drawing represents 100 measuring units on the product.
C An incline slope of 1:100.
D A decline slope of 1:100.

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1.14 The purpose of a radiant-heat system is to …

A heat water while it circulates around the combustion chamber of a boiler.
B provide a comfortable environment by means of controlling surface temperatures and minimise excessive air movement.
C allow for one or more heaters and one thermostat per room.
D pump forced air by means of a compressor.

1.15 A crane driver must …

A examine the crane ropes and slings at the start of a shift.
B let the load hang uncontrolled.
C swing a load over the heads of workers.
D exceed the maximum load limit for a specific lifting position of the jib.

(15 × 1) **[15]**

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (2.1–2.9) in the ANSWER BOOK. Also give ONE explanation for each answer.

2.1 During hacksaw work, the blade only cuts during the forward stroke.

2.2 When using the protractor head of the combination set, the incline of a work piece can be determined in degrees.

2.3 Hacksaw work does not classify as a shaping technique.

2.4 A welder and his/her assistant must provide their own personal protective equipment during work-related periods.

2.5 A sand bucket can replace a CO_2 fire extinguisher if there is an electrical fire in the work shop.

2.6 Hot cracks due to welding occur after the metal has cooled down.

2.7 In commercial applications, it is very important to have ventilation.

2.8 For installation purposes, components can be marked with any mark.

2.9 In a Lang's lay rope, the wires in the strands lie in one direction whereas the strands lie in the opposite direction.

(9 × 2) **[18]**

QUESTION 3

3.1 State the TWO commercial oxides/iron ores used to manufacture iron. (2)

3.2 State THREE quantities of elements iron absorbs during the melting process when manufacturing iron. (3)

3.3 State the use of each of the following metals:

3.3.1 White cast iron

3.3.2 Brass

3.3.3 Medium carbon steel

(3 × 1) (3)

3.4 Describe how a hammer head is secured to a handle. (2)

3.5 Describe the meaning of the code '32T' with regard to hacksaw blades. (1)

3.6 List THREE types of cuts found on files. (3)

3.7 FIGURE 1 illustrates a certain process performed with a specific hand tool. Describe the process in short.

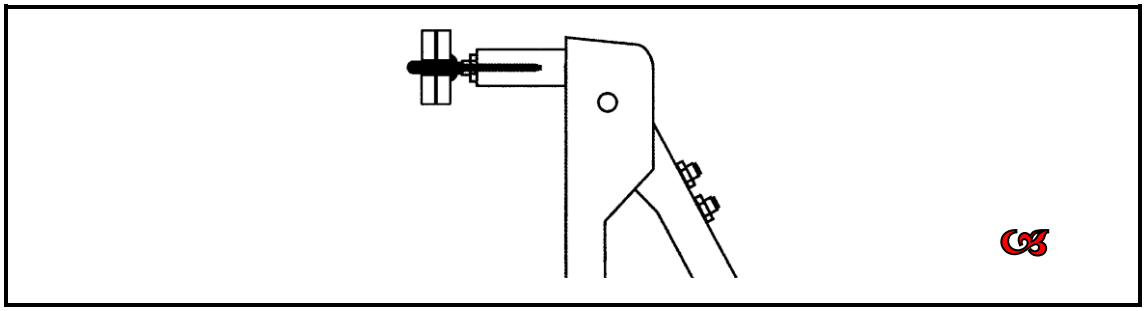


FIGURE 1 (2)

3.8 Name the tool and state the function of the tool shown in FIGURE 2.

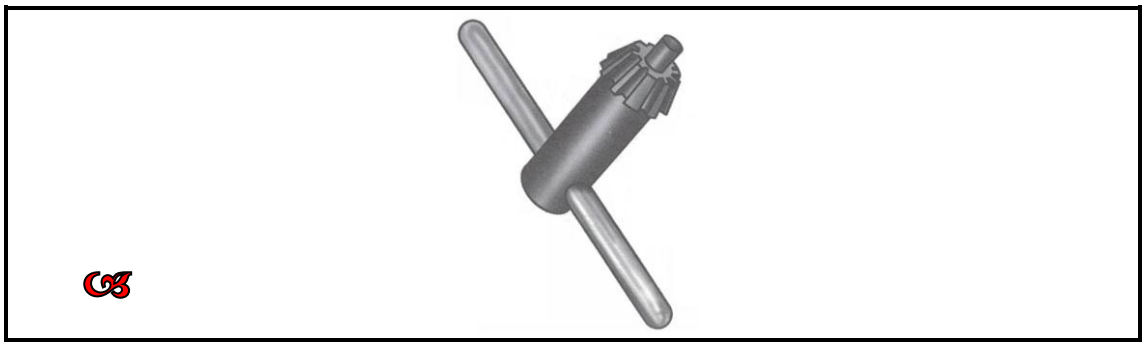


FIGURE 2 (2)

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3.9 When connecting an electrical plug on a single-phase electrical cord, you will find three electrical cores inside the cord. Re-draw the following table in your ANSWER BOOK and complete the information needed to identify the cores correctly. Ensure that the information corresponds accurately.

COLOR OF CORE	PURPOSE OF CORE

(6)

3.10 The term *clearance* refers to a certain setting on the guillotine machine regarding the blades.

Explain what is meant and say why it is necessary. (2)

3.11 Name the TWO systems in which oxy-acetylene welding can be done. (2)

3.12 Describe the purpose of the blowpipe that forms part of the oxy-acetylene gas torch. (1)

3.13 To be able to determine the correct bending allowance on plates, you have to take certain sizes in consideration.

State TWO of these sizes to consider when bending a plate. (2)

3.14 Describe the purpose of a *thermostat* with regard to HVAC systems. (2)

3.15 Sound or noise levels are measured in decibel units. According to the OH&S Act, a worker must use earmuffs from a certain decibel level to prevent hearing loss.

From which decibel level must earmuffs be used in the case of high noise levels? (For example, 20 dB) (1)

[34]

QUESTION 4

4.1 List FOUR main categories of steel. (4)

4.2 Name TWO functions of each type of steel mentioned in QUESTION 4.1. (8)

4.3 FIGURE 3 indicates the front and left views of an Equal Square to Square T-pipe connection in first-angle orthographic projection. Construct to good proportion by using the correct development method the TRUE SHAPE of the hole in the main pipe, making use of the given dimensions and corner indications. It is not necessary to redraw the given views.

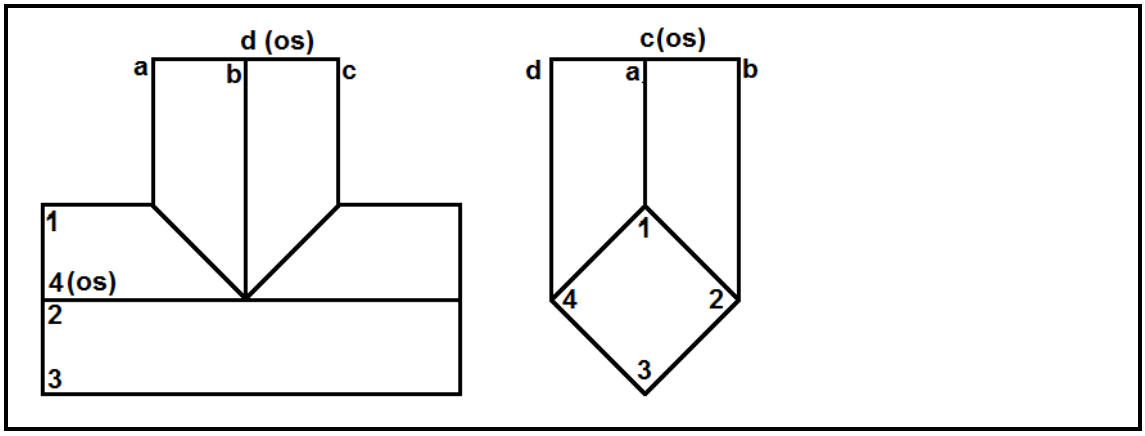


FIGURE 3 (6)

4.4 Make a neat drawing to good proportion illustrating two 45° notches cut into a plate which enable the plate to have two 90° bends. Clearly indicate the 45° angles on each notch. (4)

4.5 Which cutting process makes use of high-velocity, pressurized water and a mixture of abrasive substances to cut through steel? (1)

4.6 What does the abbreviation MSDS stand for with regard to the safe handling of substances such as fluxes? (1)

4.7 To obtain the required contour for a welding bead, certain processes can be applied such as machining. This process is indicated by the letter 'M'.

Which process is indicated by the letter 'G'? (1)

4.8 Why is it necessary to keep gas cylinders cool and not to expose them to any source of heat? (1)

4.9 Why must the cylinder valve of a gas cylinder always be kept closed when empty? (1)

4.10 Indicate whether the following statement is TRUE or FALSE. Write only 'True' or 'False' next to the question number (4.10) in the ANSWER BOOK.

Oxygen cylinders may be stacked horizontally provided that they are firmly secured at each end to prevent them from rolling. (1)

4.11 Inspections on cranes include inspections of their mechanical parts, wire ropes, chain slings and so on. Inspections must be done at specific intervals.

How often must inspections be done on mechanical parts? (1)

4.12 Steel ropes with *wire cores* have certain characteristics which make them very useful.

Describe the TWO main characteristics of this type of rope. (2)

4.13 When inspecting a steel rope, its condition should be recorded in a log book.

State TWO points that need to be reported. (2)

[33]

TOTAL: 100

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